

Exam Date & Time: 08-Jan-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (CSE/IT) EXAMINATIONS - JANUARY 2024
 SUBJECT: MAT 2126/MAT_2126/MAT 2155 - ENGINEERING MATHEMATICS - III
 ENGINEERING MATHEMATICS - III [MAT 2126]

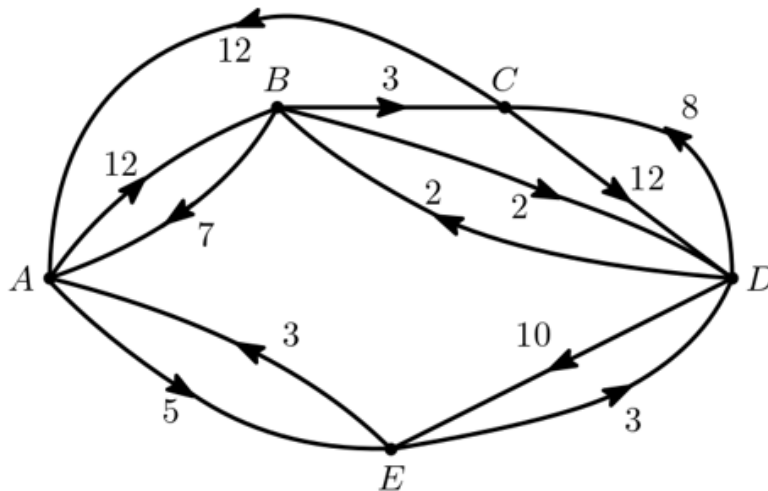
Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Consider the complete graph on 8 vertices and the cycle graph with 5 vertices. For each of these graphs, state with justification whether the graph is
 i) self-complementary
 ii) bipartite. (3)
- 1B) Prove that a graph is a tree if and only if between every pair of vertices there exists a unique path. (3)
- 1C) i) Show that for all $n > 3$, the complement of the path graph P_n is connected. (4)
 ii) Show that $\neg(P \leftrightarrow Q) \Leftrightarrow (P \leftrightarrow \neg Q)$.
- 2A) Using Dijkstra's algorithm, find shortest paths from A to all the other vertices in the graph given below. (3)



- 2B) Show that $(Z_7 - \{0\}, \times_7)$ forms an abelian group under the operation of multiplication modulo 7. (3)
- 2C) Let H be a subgroup of G . Then show that H is a normal subgroup of G if and only if $xH = Hx, \forall x \in G$. Give an example of a normal subgroup of G . (4)
- 3A) Prove that every subgroup of a cyclic group is cyclic. Illustrate this result with an example. (3)
- 3B) Given integers 1, 2, ..., 11. Two groups are made, the first group contains 5 integers and the second group contains 2 integers. In how many ways can the selection be made with no repetition if
 i) There are no further restrictions
 ii) A group has either only odd integers or only even integers. (3)
- 3C) In how many ways can 20 identical objects be distributed into 5 distinct boxes such that the first box can have at most 3 objects and the second box can have at most 7 objects? (4)

- 4A) Show that the join and meet operations in a lattice are associative. (3)
- 4B) Let $E(x_1, x_2, x_3) = \overline{((x_1 \wedge x_2) \vee (\overline{x_2} \wedge x_3))}$ be a Boolean Expression over the two-valued Boolean Algebra. Write $E(x_1, x_2, x_3)$ in both DNF and CNF. (3)
- 4C) Find the following permutations for $n = 5$ with initial permutation 12345.
 i) 97^{th} permutation in Lexicographical order
 ii) 108^{th} permutation in Fike's order (4)
- 5A) Show that $A \rightarrow \neg D$ follows logically from the premises $A \rightarrow (B \vee C)$, $B \rightarrow \neg A$ and $C \rightarrow \neg D$. (3)
- 5B) Show that the conclusion $(\exists x)[A(x) \rightarrow \neg B(x)]$ follows logically from $(\exists y)[P(y) \wedge \neg Q(y)]$ and $(\forall x)[A(x) \wedge B(x)] \rightarrow (\forall y)[P(y) \rightarrow Q(y)]$. (3)
- 5C) i) In a Boolean algebra, for any two elements a and b show that
 (a) $\overline{a \vee b} = \overline{a} \wedge \overline{b}$ (b) $\overline{a \wedge b} = \overline{a} \vee \overline{b}$ (4)
 ii) Prove that if the meet operation is distributive over the join operation in a lattice, then the join operation is distributive over the meet operation.

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